```
clc;
clear;
close all;

% Experiment 2: Chebyshev Type-I IIR Filters

Fs = 10000;
Ap = 0.5;
As = 30;

% Low-Pass Filter
fp_lp = 1500;
fs_lp = 2000;

wp_lp = fp_lp/(Fs/2);
ws_lp = fs_lp/(Fs/2);

[N_lp, Wn_lp] = cheb1ord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = cheby1(N_lp, Ap, Wn_lp, 'low');

% High-Pass Filter
fp_hp = 2500;
fs_hp = 1800;

wp_hp = fp_hp/(Fs/2);
ws_hp = fs_hp/(Fs/2);

[N_hp, Wn_hp] = cheb1ord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = cheby1(N_hp, Ap, Wn_hp, 'high');

% Display Results
fprintf('CHEBYSHEV TYPE-I IIR FILTER DESIGN\n\n');
```

```
CHEBYSHEV TYPE-I IIR FILTER DESIGN
```

```
fprintf('LOW-PASS FILTER\n');
```

```
LOW-PASS FILTER
```

```
fprintf('Filter Order = %d\n', N_lp);
```

```
Filter Order = 6
```

```
fprintf('Cutoff Frequency = %.2f Hz\n', Wn_lp*Fs/2);
```

```
Cutoff Frequency = 1500.00 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_lp);
```

```
    0.0006    0.0038    0.0095    0.0127    0.0095    0.0038    0.0006
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_lp);
```

```
    1.0000   -3.7650    6.8198   -7.3274    4.8717   -1.8927    0.3368
```

```
fprintf('\nHIGH-PASS FILTER\n');
```

```
HIGH-PASS FILTER
```

```
fprintf('Filter Order = %d\n', N_hp);
```

```
Filter Order = 6
```

```
fprintf('Cutoff Frequency = %.2f Hz\n', Wn_hp*Fs/2);
```

```
Cutoff Frequency = 2500.00 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_hp);
```

```
    0.0117   -0.0704    0.1761   -0.2348    0.1761   -0.0704    0.0117
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_hp);
```

```
    1.0000    1.3568    2.1347    1.8511    1.2978    0.5937    0.1650
```

```
% Frequency Response
[H_lp, f] = freqz(b_lp, a_lp, 2048, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 2048, Fs);

% Plot Responses
figure('Name','Chebyshev Type-I IIR Filters');

subplot(2,2,1);
plot(f, 20*log10(max(abs(H_lp),1e-8)), 'b', 'LineWidth',1.5);
grid on;
title('Low-Pass Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);
```

```
xlim([0 Fs/2]);

subplot(2,2,2);
plot(f, unwrap(angle(H_lp))*180/pi, 'r', 'LineWidth',1.5);
grid on;
title('Low-Pass Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);

subplot(2,2,3);
plot(f, 20*log10(max(abs(H_hp),1e-8)), 'b', 'LineWidth',1.5);
grid on;
title('High-Pass Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);
xlim([0 Fs/2]);

subplot(2,2,4);
plot(f, unwrap(angle(H_hp))*180/pi, 'r', 'LineWidth',1.5);
grid on;
title('High-Pass Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);
```

